ACS MIDTERM CRIB SHEET ˙w˙

LOOK UP / Laplace Table:

	$f(t)$	$F(s)$	
1	$k \cdot f(t)$	$kF(s)$	
2	$f_1(t)+f_2(t)$	$F_1(s)+F_2(s)$	
3	$\delta(t)$	1	impulse
4	$1(t)$	$\frac{1}{s}$	set point
5	t	$\frac{1}{s^2}$	ramp
6	$\frac{t^2}{2}$	$\frac{1}{s^3}$	
7	t^n	$\frac{n!}{s^{n+1}}$	$\frac{2}{s^3}$ parabola
8	$\frac{t^{n-1}}{(n-1)!}$	$\frac{1}{s^n}$	
9	e^{-at}	$\frac{1}{s+a}$	exponential
10	$\sin(bt)$	$\frac{b}{s^2+b^2}$	
11	$\cos(bt)$	$\frac{s}{s^2+b^2}$	
12	$e^{-at}\sin(bt)$	$\frac{b}{(s+a)^2+b^2}$	
13	$e^{-at}\cos(bt)$	$\frac{s+a}{(s+a)^2+b^2}$	
14	$t^n e^{-at}$	$\frac{n!}{(s+a)^{n+1}}$	
15	$\dot{f}(t)$	$sF(s)-f(0)$	
16	$\frac{d^n f(t)}{dt^n}$	$s^nF(s)+\text{initial conditions}$	
17	$\int f(t)dt$	$\frac{F(s)}{s}$	
18	$\int \cdots \int f(t)dt$	$\frac{F(s)}{s^n}$	
19	$f_1(t) * f_2(t)$	$F_1(s) \cdot F_2(s)$	
20	$f_1(t) \cdot f_2(t)$	$F_1(s) * F_2(s)$	
21	$\lim_{t\to 0} f(t)$	$y(0) = \lim_{s\to\infty} sF(s)$	IVT
22	$\lim_{t\to\infty} f(t)$	$y(\infty) = \lim_{s\to 0} sF(s)$	FVT

open-loop ⟶ closed loop:

closed loop: $\frac{G}{1+GH}$

3 models ⇒

1] strictly proper ⇒ $\frac{B}{A} = \frac{1}{A/B} = \frac{1}{1/G + H}$ H is simple, G is the rest

ex) $\frac{Y}{R} = T = \frac{s+4}{s^2+s+9} \Rightarrow \frac{1}{\frac{s^2+s+9}{s+4}} = \frac{1}{\frac{s^2+s}{s+4} + \frac{9}{s+4}}$

loop gain:

$H = \frac{9}{s+4}$, $G = \frac{s+4}{s^2+s} \Rightarrow L = GH = \left(\frac{\cancel{s+4}}{s^2+s}\right)\left(\frac{9}{\cancel{s+4}}\right)$

$L = \frac{9}{s^2+s}$ //

2] biproper ⇒ long division to factor out constant

ex) $\frac{Y}{R} = T = \frac{B}{A} = \frac{s^2+3s+9}{s^2-2s-7} = 1 + \frac{5s+16}{s^2-2s-7}$

loop gain w/ this

$$\begin{array}{r} 1 \\ s^2-2s-7 \overline{) s^2+3s+9} \\ -\underline{s^2-2s-7} \\ 5s+16 \end{array}$$

3] improper ⇒ don't even think about it

ex) $\frac{s^3+s^2+9s+9}{s^2+4s+2}$

BIBO Stability:

$\frac{Y}{U} = \frac{B}{A}$ $A = \emptyset$, $s = \begin{Bmatrix} \text{determine} \\ \text{BIBO} \\ \text{stability} \end{Bmatrix}$

$s = \sigma + j\omega$

only real value (σ) determines

1] $\sigma < \emptyset$, system is stable

2] $\sigma > \emptyset$, system is unstable

3] $\sigma = \emptyset$, marginally stable

(only cancel stable poles)

☆ $y(\infty)$ … ☆ (IVT)

denom ↑ ∴ $= \emptyset$

dem = num ∴ $= \frac{\square s^{max}}{\square s^{max}}$

num ↑ ∴ $= \infty$

$\frac{Y}{U} = \frac{s+3}{s^2+3s+2} = \frac{s+3}{(s+2)(s+1)} = \frac{A}{s+2} + \frac{B}{s+1}$

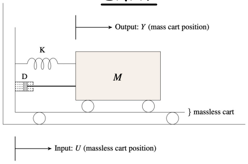
A: $\left(\frac{s+3}{s+1}\right)\Big|_{s=-2} = -1$

B: $\left(\frac{s+3}{s+2}\right)\Big|_{s=-1} = 2$

$\frac{-1}{s+2} + \frac{2}{s+1}$

Practical Examples:

CART



To find the open-loop transfer function from the free body diagram, we start with Newton's second law

$$\sum F = ma \tag{5.1}$$

where $\sum F$ is the summation of all of the forces acting on the system, m is the mass of the mass cart and a is the acceleration of the mass cart. Since $y(t)$ is defined as the position of the mass cart, the acceleration of the mass cart is the second derivative of the cart's position

$$a = \ddot{y}(t)$$

and

$$\sum F = F_D + F_S$$

where F_D is the damper force

$$F_D = -D(\dot{y}(t) - \dot{u}(t))$$

and F_S is the spring force

$$F_S = -K(y(t) - u(t))$$

Newton's second law for the free body diagram is

$$m\ddot{y}(t) = -D(\dot{y}(t) - \dot{u}(t)) - K(y(t) - u(t))$$

$$m\ddot{y}(t) = -D\dot{y}(t) + D\dot{u}(t) - Ky(t) + Ku(t)$$

transform into the Laplace domain

$$ms^2Y(s) = -DsY(s) + DsU(s) - KY(s) + KU(s)$$

isolate the I/O signals

$$ms^2Y(s) + DsY(s) + KY(s) = DsU(s) + KU(s)$$

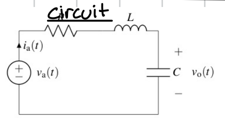
and factor them

$$(ms^2 + Ds + K)Y(s) = (Ds + K)U(s)$$

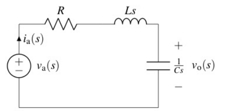
Finally, write the open-loop transfer function

$$\frac{Y(s)}{U(s)} = \frac{Ds + K}{ms^2 + Ds + K}$$

Circuit



Let's transform the circuit into the Laplace Domain equivalent



Perform KVL around the circuit

$$-V_a(s) + Ri_a(s) + Lsi_a(s) + \frac{i_a(s)}{Cs} = 0$$

Now, define the current through the circuit with respect to the output signal using Ohm's Law

$$i_a(s) = \frac{V_o(s)}{\frac{1}{Cs}} = Csv_o(s)$$

the KVL equation simplifies to

$$-V_a(s) + RCsV_o(s) + LCs^2V_o(s) + V_o(s) = 0$$

Since a transfer function is a ratio of output to input, we need to isolate the I/O signals

$$V_a(s) = RCsV_o(s) + LCs^2V_o(s) + V_o(s)$$

$$V_a(s) = (RCs + LCs^2 + 1)V_o(s)$$

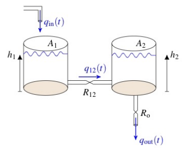
$$\frac{V_o(s)}{V_a(s)} = \frac{1}{RCs + LCs^2 + 1}$$

Finally, in proper form

$$\frac{V_o(s)}{V_a(s)} = \frac{1}{LCs^2 + RCs + 1}$$

Note: "Proper form" implies the numerator and denominator are written in descending order polynomial form.

TANK



From the diagram, determine the fluid flow from input to output and construct an open-loop transfer function. First, define the differential equations. The volume of fluid in the first tank is

$$V_1(t) = A_1 \cdot h_1(t)$$

however, there is fluid flowing into the tank, $q_{in}(t)$ and fluid flowing out of the tank, $q_{12}(t)$; so we consider the rate of change of the fluid in the first tank.

$$q_{in}(t) - q_{12}(t) = \frac{dV_1(t)}{dt}$$

or written with h_1 as the dynamic variable

$$q_{in}(t) - q_{12}(t) = A_1 \cdot \frac{dh_1(t)}{dt} \tag{5.2}$$

The flow from tank 1 to tank 2 is a function of the heights of the tanks and the restriction of the flow tube.

$$q_{12}(t) = \frac{h_1(t) - h_2(t)}{R_{12}} \tag{5.3}$$

where $1 \le R_{12} < \infty$, the larger R_{12} the greater the restriction on the flow rate.

Similar to the fluid flow in tank 1, the fluid flow and change of volume of tank 2 is

$$q_{12}(t) - q_{out}(t) = A_2 \cdot \frac{dh_2(t)}{dt} \tag{5.4}$$

Finally, the output equation is simply

$$q_{out}(t) = \frac{h_2(t)}{R_o} \tag{5.5}$$

where $1 \le R_o < \infty$, the larger R_o the greater the restriction on the flow rate.

Now that we have defined the system from input to output we need to convert (5.2) to (5.5) into the Laplace domain equivalent equations

Starting with (5.2)

$$Q_{in}(s) - Q_{12}(s) = A_1sH_1(s)$$

now, we rewrite it with the input $Q_{in}(s)$ and the output $H_1(s)$

$$H_1(s) = \frac{Q_{in}(s) - Q_{12}(s)}{A_1s} \tag{5.6}$$

Moving to (5.3)

$$Q_{12}(s) = \frac{H_1(s) - H_2(s)}{R_{12}} \tag{5.7}$$

Next, (5.4)

$$Q_{in}(s) - Q_{12}(s) = A_1sH_1(s)$$

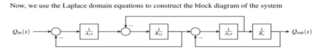
Now, we rewrite the equation with the input $Q_{in}(s)$ and the output $H_1(s)$

$$H_2(s) = \frac{Q_{12}(s) - Q_{out}(s)}{A_2s} \tag{5.8}$$

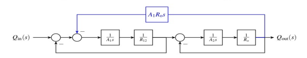
Finally (5.5)

$$Q_{out}(s) = \frac{H_2(s)}{R_o} \tag{5.9}$$

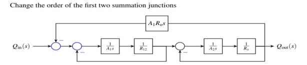
Now, we use the Laplace domain equations to construct the block diagram of the system



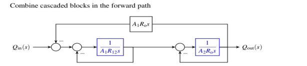
Next, we need to simplify the block diagram down to a transfer function. First move the branch from in front of the $\frac{1}{R_o}$ to after it and move the summation junction after $\frac{1}{A_1s}$ to before it



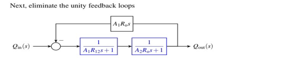
Change the order of the first two summation junctions



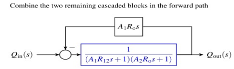
Combine cascaded blocks in the forward path



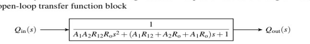
Next, eliminate the unity feedback loops



Combine the two remaining cascaded blocks in the forward path

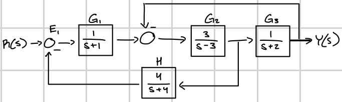


Finally, eliminate the remaining feedback loop and reduce the diagram to produce a single open-loop transfer function block



Of course, the single open-loop block diagram reveals the input/output transfer function

$$\frac{Q_{out}(s)}{Q_{in}(s)} = \frac{1}{A_1A_2R_{12}R_os^2 + (A_1R_{12} + A_2R_o + A_1R_o)s + 1}$$



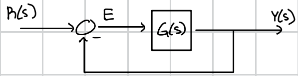
$$Y(s) = E_2G_2G_3$$
$$E_2 = G_1E_1 - Y(s)$$
$$E_1 = R(s) - E_2G_2H$$

$$E_2 = G_1(R - E_2G_2H) - Y$$
$$E_2 = G_1R - G_1G_2HE_2 - Y$$
$$E_2 + G_1G_2HE_2 = G_1R - Y$$
$$E_2(1 + G_1G_2H) = G_1R - Y$$

$$Y = \frac{(G_1R - Y)G_2G_3}{1 + G_1G_2H} \qquad E_2 = \frac{G_1R - Y}{1 + G_1G_2H}$$
$$Y + YG_1G_2H = G_1G_2G_3R - G_2G_3Y$$
$$Y + YG_1G_2H + G_2G_3Y = G_1G_2G_3R$$
$$Y(1 + G_1G_2H + G_2G_3) = (G_1G_2G_3)R$$
$$\frac{Y}{R} = \frac{G_1G_2G_3}{1 + G_1G_2H + G_2G_3} = \frac{G_1G_2G_3}{1 + G_2(G_1H + G_3)}$$
$$= \frac{\left(\frac{1}{s+1}\right)\left(\frac{3}{s-3}\right)\left(\frac{1}{s+2}\right)}{1 + \left(\frac{3}{s-3}\right)\left[\left(\frac{1}{s+1}\right)\left(\frac{4}{s+4}\right) + \left(\frac{1}{s+2}\right)\right]} = \text{calculator...}$$
$$= \boxed{\frac{3s+12}{s^4+4s^3-4s^2-7s+12}}$$

loop gain! $L = GH$ $\qquad \frac{B}{A} = \frac{1}{A/B} = \frac{1}{1/G + H}$

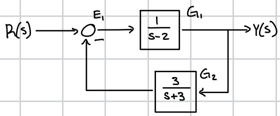
$$\frac{3s+12}{s^4+4s^3-4s^2-7s+12} = \frac{1}{\frac{s^4+4s^3-4s^2-7s+12}{3s+12}}$$
$$= \frac{1}{\underbrace{\frac{s^4+4s^3-4s^2-7s}{3s+12}}_{1/G} + \underbrace{\frac{12}{3s+12}}_{H}}$$
$$H = \frac{12}{3s+12}, \quad G = \frac{3s+12}{s^4+4s^3-4s^2-7s} \quad \Big\} \quad L = GH = \left(\frac{\cancel{3s+12}}{s^4+4s^3-4s^2-7s}\right)\left(\frac{12}{\cancel{3s+12}}\right)$$
$$\boxed{L = \frac{12}{s^4+4s^3-4s^2-7s}}$$



$$G(s) = 2\ddot{y}(t) - 2\dot{e}(t) = -4e(t) - 3y(t)$$
$$2\ddot{y}(t) + 3y(t) = 2\dot{e}(t) - 4e(t)$$
$$2s^2Y(s) + 3Y(s) = 2sE(s) - 4E(s)$$
$$2s^2Y(s) + 3Y(s) = 2s(R(s) - Y(s)) - 4(R(s) - Y(s))$$
$$= 2sR(s) - 2sY(s) - 4R(s) - 4Y(s)$$
$$Y(s)\left[2s^2+2s-1\right] = R(s)\left[2s-4\right]$$
$$\boxed{\frac{Y(s)}{R(s)} = \frac{2s-4}{2s^2+2s-1}} = \frac{B}{A}$$

loop gain! $L = GH$

$$\frac{1}{A/B} = \frac{1}{\frac{2s^2+2s-1}{2s-4}} = \frac{1}{\underbrace{\frac{2s^2+2s}{2s-4}}_{1/G} - \underbrace{\frac{1}{2s-4}}_{H}}$$
$$H = -\frac{1}{2s-4}, \quad G = \frac{2s-4}{2s^2+2s} \quad \Big\} \quad L = \left(\frac{\cancel{2s-4}}{2s^2+2s}\right)\left(\frac{-1}{\cancel{2s-4}}\right) = \boxed{-\frac{1}{2s^2+2s}}$$



differential equation of the system for a generic input, $r(t)$

$$Y(s) = G_1E_1, \quad E_1 = R(s) - G_2Y(s) \quad \Big\} \quad Y = G_1(R - G_2Y)$$
$$Y = G_1R - G_1G_2Y$$
$$Y + G_1G_2Y = G_1R$$
$$Y(1 + G_1G_2) = (G_1)R$$
$$\frac{Y}{R} = \frac{G_1}{1+G_1G_2} = \frac{\left(\frac{1}{s-2}\right)}{1 + \left(\frac{1}{s-2}\right)\left(\frac{3}{s+3}\right)} = \frac{s+3}{s^2+s-3}$$
$$\frac{Y}{R} = \frac{s+3}{s^2+s-3}$$
$$Y(s^2+s-3) = R(s+3)$$
$$Ys^2 + Ys - 3Y = Rs + 3R$$
$$\boxed{\ddot{y}(t) + \dot{y}(t) - 3y(t) = \dot{r}(t) + 3r(t)}$$

BIBO stable? $\frac{Y}{R} = \frac{s+3}{s^2+s-3}$, $\quad s = \frac{-1 \pm \sqrt{13}}{2} = -2.3, +1.3$

not BIBO stable b.c. poles need to have negative real parts!

$$\frac{Y}{R} = \frac{s^2+37s+36}{(s+12)(s^2+8s+7)} \qquad \text{input is setpoint}$$

$$\text{Setpoint} = \frac{1}{s} = R(s)$$
$$Y(s) = R(s)\frac{s^2+37s+36}{(s+12)(s^2+8s+7)} = \frac{1}{s}\left(\frac{s^2+37s+36}{(s+12)(s^2+8s+7)}\right)$$
$$y(\infty) = \lim_{s\to 0} s\cdot Y(s) = \lim_{s\to 0} \cancel{s}\,\frac{1}{\cancel{s}}\left(\frac{s^2+37s+36}{(s+12)(s^2+8s+7)}\right)$$
$$= \frac{0^2+0+36}{(0+12)(0^2+0+7)} = \frac{36}{12\cdot 7} = \boxed{\frac{3}{7}}$$

$$\frac{Y}{R} = \frac{B}{A} \Rightarrow A = (s+12)(s^2+8s+7)$$
$$= s^3 + 20s^2 + 103s + 84$$

no sign changes
∴ no unstable poles

s^3	1	103
s^2	20	84
s^1	98.8	0
s^0	84	

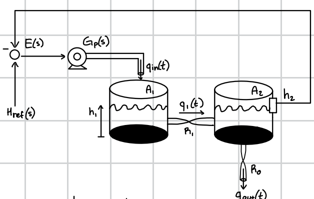
$$x_{[3,1]} = \begin{vmatrix} 1 & 103 \\ 20 & 84 \end{vmatrix}$$
$$= \frac{(20)(103) - (1)(84)}{20}$$
$$= 98.8$$

$$y(0) = \lim_{s\to\infty} s\cdot Y(s) = \lim_{s\to\infty} \cancel{s}\,\frac{1}{\cancel{s}}\left(\frac{s^2+37s+36}{(s+12)(s^2+8s+7)}\right) = \boxed{0}$$

Given: $V(s) = \frac{400}{s^2+8s+400} = \frac{400}{(s^2+8s+16)+384}$

↳ F: $v(t)$

$$384 = b^2 \Rightarrow b = \sqrt{384} = 8\sqrt{6}$$
$$s = -4 \pm 8\sqrt{6}\,j$$
$$\frac{b}{(s+a)^2+b^2} = \frac{400}{(s^2+8s+16)+8\sqrt{6}} \cdot \frac{8\sqrt{6}}{8\sqrt{6}}$$
$$\Big\} \ \mathcal{L}^{-1}$$
$$\frac{400}{8\sqrt{6}}e^{-4t}\sin(8\sqrt{6}\,t) = \frac{50}{\sqrt{6}}e^{-4t}\sin(8\sqrt{6}\,t)$$



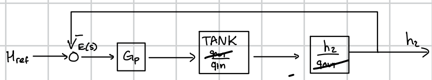
Parameter	Value	Identification
h_2		height of tank 2
h_{ref}		reference height
A_1	2	area tank 1
A_2	3	area tank 2
R_1	1	restriction $q_1(t)$
R_o	1	restriction $q_{out}(t)$
$G_p(s)$	$\frac{1}{s+1}$	pump

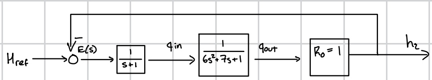
F: transfer function $\frac{H_2(s)}{H_{ref}(s)}$

$$q_{out} = \frac{h_2}{R_o} \Rightarrow R_o = \frac{h_2}{q_{out}}$$

$$\frac{q_{out}}{q_{in}} = \frac{1}{A_1A_2R_{12}R_os^2 + (A_1R_{12} + A_2R_o + A_1R_o)s + 1} \quad \Big\} \text{notes}$$

$$\frac{q_{out}}{q_{in}} = \frac{1}{(2)(3)(1)s^2 + (2\cdot 1 + 3\cdot 1 + 2\cdot 1)s + 1}$$
$$= \frac{1}{6s^2+7s+1}$$



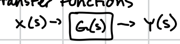


$$h_2 = R_oTG_pE, \quad E = H_{ref} - h_2 \quad \Big\} \quad h_2 = R_oTG_p(H_{ref} - h_2)$$
$$h_2 = R_oTG_pH_{ref} - R_oTG_ph_2$$
$$h_2 + R_oTG_ph_2 = R_oTG_pH_{ref}$$
$$h_2(1 + R_oTG_p) = (R_oTG_p)H_{ref}$$

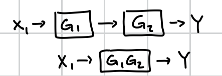
$$\frac{h_2}{H_{ref}} = \frac{R_oTG_p}{1 + R_oTG_p} = \frac{\frac{1}{(6s^2+7s+1)(s+1)}}{1 + \left(\frac{1}{(6s^2+7s+1)(s+1)}\right)} = \boxed{\frac{1}{6s^3+13s^2+8s+2}}$$

Block Diagrams:

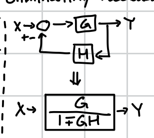
transfer functions



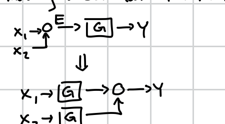
$$\frac{Y(s)}{X(s)} = G(s)$$



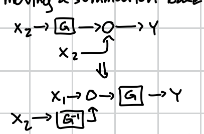
eliminating feedback



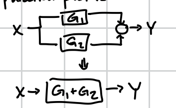
moving a summation forward



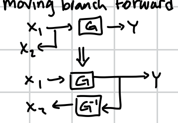
moving a summation back



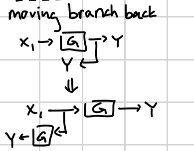
parallel paths



moving branch forward



moving branch back



Kira Giannetti 2170909 SPRING 2025

ACS FINAL CRIB SHEET

steady state errors

input	$R(s)$		Formula
step	$\frac{1}{s}$	$K_p = \lim_{s\to 0} G(s)H(s)$	$e_{ss} = \frac{1}{1+K_p}$
ramp	$\frac{1}{s^2}$	$K_v = \lim_{s\to 0} sG(s)H(s)$	$e_{ss} = \frac{1}{K_v}$
parabola	$\frac{1}{s^3}$	$K_a = \lim_{s\to 0} s^2G(s)H(s)$	$e_{ss} = \frac{1}{K_a}$

PI

$$G_c = K_p + \frac{K_i}{s} = \frac{K_p(s+\hat{K}_i)}{s}$$

$$\hat{K}_i = \frac{K_i}{K_p}$$

[pick one zero for $(s+\hat{K}_i)$, increase system type]

PID

$$G_c = K_p + \frac{K_i}{s} + K_d s = \frac{K_d(s^2 + \hat{K}_p s + \hat{K}_i)}{s} = \frac{K_d(s+\beta_1)(s+\beta_2)}{s}$$

$$\hat{K}_i = \frac{K_i}{K_d}, \quad \hat{K}_p = \frac{K_p}{K_d}$$

[pick 2 zeros for $(s+\beta_1)$ and $(s+\beta_2)$, increase system type]

PD

$$G_c = K_p + K_d s \rightarrow K_d(s+\hat{K}_p), \quad \hat{K}_p = \frac{K_p}{K_d}$$

[Choose one zero at $(s+\hat{K}_p)$, does not increase system type]

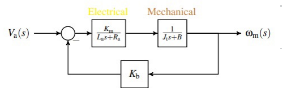
I

$$G_c = \frac{K_i}{s}$$

[increase system type]

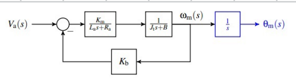
DC Servo Motors

General:

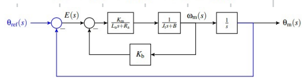


$$\frac{\omega_m(s)}{V_a(s)} = G_m(s) = \frac{K_m}{L_a J_t s^2 + (L_a B + R_a J_t)s + R_a B + K_m K_b}$$

Position Tracking:

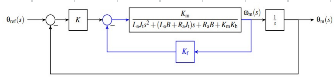


$$\frac{\theta_m(s)}{V_a(s)} = \frac{K_m}{s\left(L_a J_t s^2 + (L_a B + R_a J_t)s + R_a B + K_m K_b\right)}$$
$$= \frac{K_m}{L_a J_t s^3 + (L_a B + R_a J_t)s^2 + (R_a B + K_m K_b)s}$$



$$\frac{\theta_m(s)}{\theta_{ref}(s)} = \frac{K_m}{L_a J_t s^3 + (L_a B + R_a J_t)s^2 + (R_a B + K_m K_b)s + K_m}$$

Tachometer feedback:



$$\frac{\theta_m(s)}{\theta_{ref}(s)} = \frac{K \cdot K_m}{L_a J_t s^3 + (L_a B + R_a J_t)s^2 + (R_a B + K_m K_b + K_m K_t)s + K \cdot K_m}$$

Root Locus:

$$\sigma_i = \frac{\Sigma(\text{real part poles}) - \Sigma(\text{real part zeros})}{\#\text{ of poles} - \#\text{ of zeros}}$$

$$K = \{0, 1, \ldots, |\#\text{ of poles} - \#\text{ of zeros}| - 1\}$$

$$\Theta_k = \frac{(2k+1)\pi}{\#\text{poles} - \#\text{zeros}}$$

BA/BI: $N_L' D_L - D_L' N_L = \emptyset$

$$K_{BA} = \left.\frac{-D_L}{N_L}\right|_{s=BA}, \quad \left.\frac{-D_L'}{N_L'}\right|_{s=BA}$$

$K_{cr} \rightarrow A = D_L + K N_L$

↳ routh array and determine stability for K

position control system, DC Motor
w/ tachometer feedback, $\emptyset \le K_t < \infty$
and proportional control, $\emptyset \le K_p < \infty$

parameters:
↳ $K_m = 10,\ L_a = 0.01,\ K_b = 0.10,\ J_t = 2,\ B = 0.25,\ R_a = 6.0$

a] set $K_t = 0.25$; determine range for stable K_p
b] set $K_p = 10,\ K_t = 7$; determine
(i) open loop transfer function in poly form
(ii) zeroes and poles

a]
$$\frac{\theta_m}{\theta_{ref}} = \frac{K_p \cdot K_m}{L_a J_t s^3 + (L_a B + R_a J_t)s^2 + (R_a B + K_m K_b + K_m K_t)s + K_p K_m}$$
$$= \frac{10K_p}{(0.01)(2)s^3 + [(0.01)(0.25)+(6)(2)]s^2 + [(6)(0.25)+(10)(0.1)+(10)(0.25)] + (10)K_p}$$
$$= \frac{10K_p}{0.02s^3 + 12.0025s^2 + 5s + 10K_p}$$

RA: no sign changes for stability!!

s^3	0.02	5
s^2	12.0025	$10K_p$
s^1	$X_{[3,1]}$	$\emptyset$
s^0	$10K_p$	

$$X_{[3,1]} = \frac{(12.0025)(5) - 10K_p(0.02)}{12.0025} = \frac{60.0125 - 0.2K_p}{12.0025}$$

$10K_p > \emptyset$
$K_p > \emptyset$
$60.00125 - 0.2K_p > \emptyset$
$-0.2K_p > -60.00125 \Rightarrow K_p < \frac{60.00125}{0.2} \Rightarrow K_p < 300.006$

a] $\boxed{\emptyset < K_p < 300.006}$

b] $K_p = 10,\ K_t = 7,\ K_m = 10,\ L_a = 0.01,\ K_b = 0.10,\ J_t = 2,\ B = 0.25,\ R_a = 6.0$
$$\frac{\theta_m}{\theta_{ref}} = \frac{K_p \cdot K_m}{L_a J_t s^3 + (L_a B + R_a J_t)s^2 + (R_a B + K_m K_b + K_m K_t)s + K_p K_m}$$
$$= \frac{10(10)}{(0.01)(2)s^3 + [(0.01)(0.25)+(6)(2)]s^2 + [(6)(0.25)+(10)(0.1)+(10)(7)] + (10)(10)}$$
$$\boxed{\frac{\theta_m}{\theta_{ref}} = \frac{100}{0.02s^3 + 12.0025s^2 + 72.5s + 100}} \Leftarrow (i)$$

zeroes: $\boxed{\text{none}}$
poles: $0.02s^3 + 12.0025s^2 + 72.5s + 100 = \emptyset$
$\boxed{s = \{-594.037, -3.9657, -2.12245\}}$

Given: $\frac{Y(s)}{R(s)} = \frac{s+126}{(s+6)(s^2+10s+21)}$

a] loop gain L, and steady state error for position, velocity, and acceleration

$$L = GH: \frac{\frac{1}{s^3+16s^2+81s+126}}{s+126} \rightarrow \frac{1}{\frac{1}{G} + H}$$
$$H = \frac{126}{s+126}, \quad G = \frac{s+126}{s^3+16s^2+81s}$$
$$LG = GH = \left(\frac{126}{\cancel{s+126}}\right)\left(\frac{\cancel{s+126}}{s^3+16s^2+81s}\right)$$
$$\boxed{LG = \frac{126}{s(s^2+16s+81)}}$$
↑ Type 1 system

i) $K_p = \lim_{s\to 0} \frac{126}{s(s^2+16s+81)} = \frac{126}{\emptyset} = \infty \Rightarrow e_p(\infty) = \frac{1}{1+\infty} = \boxed{\emptyset}$

ii) $K_v = \lim_{s\to 0} \cancel{s} \cdot \frac{126}{\cancel{s}(s^2+16s+81)} = \frac{126}{81} = 1.56 \Rightarrow e_v(\infty) = \frac{1}{1.56} = \boxed{0.64}$

iii) $K_a = \lim_{s\to 0} s^{\cancel{2}} \cdot \frac{126}{\cancel{s}(s^2+16s+81)} = \frac{\emptyset}{81} = \emptyset \Rightarrow e_a(\infty) = \frac{1}{\emptyset} = \boxed{\infty}$

b] when input is impulse, find (1) $y(\emptyset)$, (2) $y(\infty)$, (3) $y(1)$

$$Y(s) = \frac{s+126}{s^3+16s^2+81s+126} \overset{\text{expand}}{=} \frac{119}{4(s+7)} - \frac{40}{(s+6)} + \frac{41}{4(s+3)}$$
$$y(t) = \frac{119}{4}e^{-7t} - 40e^{-6t} + \frac{41}{4}e^{-3t}$$

(1) $y(\emptyset) = \lim_{t\to\infty}(1)[y(t)] = \emptyset - \emptyset + \emptyset = \boxed{\emptyset}$
(2) $y(\infty) = \lim_{t\to\emptyset}(1)[y(t)] = \frac{119}{4} - 40 + \frac{41}{4} = \boxed{\emptyset}$
(3) $y(1) = \frac{119}{4}e^{-7} - 40e^{-6} + \frac{41}{4}e^{-3} = \boxed{0.438}$

Given: A typical negative feedback system

a] $G_c(s) = 1,\ G(s) = \frac{20s^3+60s^2+240s+200}{s^4+5s^3+6s^2},\ H(s) = \frac{2}{s+2}$

a] Routh Array to verify stability
$$A(s) = 1 + G_c(s)G(s)H(s) = 1 + \frac{(20s^3+60s^2+240s+200)\left(\frac{2}{s+2}\right)}{s^4+5s^3+6s^2} = 1 + \underbrace{\frac{(20s^3+60s^2+240s+200)\,2}{(s^4+5s^3+6s^2)(s+2)}}_{G_{open} = G(s)H(s)}$$
$$A(s) = \frac{s^5+7s^4+56s^3+132s^2+480s+400}{s^5+7s^4+16s^3+12s^2}$$

RA: no sign changes for stability!!

s^5	1	56	480
s^4	7	132	400
s^3	37.14	422.86	$\emptyset$
s^2	52.3	400	$\emptyset$
s^1	138.81	$\emptyset$	
s^0	400		

$\boxed{\text{no sign changes} \therefore \text{system is stable}}$

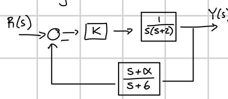
b] $OL \Rightarrow G(s)H(s) = \frac{2(20s^3+60s^2+240s+200)}{(s^4+5s^3+6s^2)(s+2)} \longrightarrow = (s+2)s^2(s^2+5s+6)$
Type 2 system

$K_p = \lim_{s\to 0} \frac{2(20s^3+60s^2+240s+200)}{s^2(s+2)(s^2+5s+6)} = \frac{400}{\emptyset} = \infty \Rightarrow e_p(\infty) = \frac{1}{1+\infty} = \boxed{\emptyset}$

$K_v = \lim_{s\to 0} \cancel{s} \cdot \frac{2(20s^3+60s^2+240s+200)}{s^{\cancel{2}}(s+2)(s^2+5s+6)} = \frac{400}{\emptyset} = \infty \Rightarrow e_v(\infty) = \frac{1}{\infty} = \boxed{\emptyset}$

$K_a = \lim_{s\to 0} \cancel{s^2} \cdot \frac{2(20s^3+60s^2+240s+200)}{\cancel{s^2}(s+2)(s^2+5s+6)} = \frac{400}{12} = 33.3 \Rightarrow e_a(\infty) = \frac{1}{33.3} = \boxed{0.03}$

control system where $\emptyset < K < \infty$

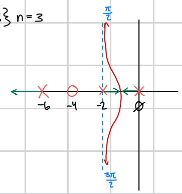


a] Root Locus for system when $\alpha = 4$
$$T = \frac{\frac{K}{s(s+2)}}{1 + \frac{K}{s(s+2)}\left(\frac{s+4}{s+6}\right)} \cdot \frac{s(s+2)(s+6)}{s(s+2)(s+6)} = \frac{K(s+6)}{s(s+2)(s+6) + K(s+4)} = \frac{K(s+6)}{s^3+8s^2+12s+K(s+4)}$$

$L = KGH = \frac{K(s+4)}{s(s+2)(s+6)}$ zeroes: $\{-4\}$ $m=1$
poles: $\{\emptyset, -2, -6\}$ $n=3$

$\sigma_i = \frac{(\emptyset-2-6)-(-4)}{3-1} = -2$

$K = \{0, 1\}$ $\theta_k = \begin{cases} \theta_0 = \frac{\pi}{2} \\ \theta_1 = \frac{3\pi}{2} \end{cases}$



BA: $K = 1,\ N_L' D_L - D_L' N_L = \emptyset$
$(1)(s^3+8s^2+12s) - (3s^2+16s+12)(s+4) = \emptyset$
$s = -1.07$

b] Range of K where system stable w/ $\alpha = 4$
$A = D_L + KN_L$
$= [s^3+8s^2+12s] + K[s+4]$
$= s^3+8s^2+(K+12)s+4K$

RA:

s^3	1	$(K+12)$
s^2	8	$4K$
s^1	$X_{[3,1]}$	$\emptyset$
s^0	$4K$	

$X_{[3,1]} = \frac{(8)(K+12)-(4K)}{8} = \frac{4K-96}{8}$

$\frac{4K-96}{8} > \emptyset$ $4K > \emptyset$
$4K - 96 > \emptyset$ $K > \emptyset$
$K > -24$

$\boxed{\text{system is stable when } K > \emptyset}$

c] Range of α where system is stable w/ $K = 10$
$A = D_L + KN_L$
$= [s^3+8s^2+12s] + [10][s+\alpha]$
$= s^3+8s^2+22s+10\alpha$

RA:

s^3	1	22
s^2	8	10α
s^1	$X_{[3,1]}$	$\emptyset$
s^0	10α	

$X_{[3,1]} = \frac{(8)(22)-10\alpha}{8} = \frac{176-10\alpha}{8}$

$\frac{176-10\alpha}{8} > \emptyset$ $10\alpha > \emptyset$
$176 - 10\alpha > \emptyset$ $\alpha > \emptyset$
$\alpha < 17.6$

$\boxed{\text{system is stable when } \emptyset < \alpha < 17.6}$

position control system, DC Motor

w/ tachometer feedback, $\emptyset \leq K_f < \infty$

and proportional control, $\emptyset \leq K_p < \infty$

parameters:

↳ $K_m = 10$, $L_a = 0.01$, $K_b = 0.10$, $J_t = 2$, $B = 0.25$, $R_a = 6.0$

a] set $K_f = 0.25$; determine range for stable K_p

b] set $K_p = 10$, $K_f = 7$; determine

(i) open loop transfer function in poly form

(ii) zeroes and poles

a]

$$\frac{\theta_m}{\theta_{ref}} = \frac{K_p \cdot K_m}{L_a J_t s^3 + (L_a B + R_a J_t)s^2 + (R_a B + K_m K_b + K_m K_f)s + K_p K_m}$$

$$= \frac{10K_p}{(0.01)(2)s^3 + [(0.01)(0.25) + (6)(2)]s^2 + [(6)(0.25) + (10)(0.1) + (10)(0.25)] + (10)K_p}$$

$$= \frac{10K_p}{0.02s^3 + 12.0025s^2 + 5s + 10K_p}$$

RA: no sign changes for stability!!

s^3	0.02	5
s^2	12.0025	$10K_p$
s^1	$X_{[3,1]}$	$\emptyset$
s^0	$10K_p$	

$$X_{[3,1]} = \frac{(12.0025)(5) - 10K_p(0.02)}{12.0025} = \frac{60.0125 - 0.2K_p}{12.0025}$$

$10K_p > \emptyset$

$K_p > \emptyset$

$60.00125 - 0.2K_p > \emptyset$

$-0.2K_p > -60.00125 \Rightarrow K_p < \frac{60.00125}{0.2} \Rightarrow K_p < 300.006$

a] $\boxed{\emptyset < K_p < 300.006}$

b] $K_p = 10$, $K_f = 7$, $K_m = 10$, $L_a = 0.01$, $K_b = 0.10$, $J_t = 2$, $B = 0.25$, $R_a = 6.0$

$$\frac{\theta_m}{\theta_{ref}} = \frac{K_p \cdot K_m}{L_a J_t s^3 + (L_a B + R_a J_t)s^2 + (R_a B + K_m K_b + K_m K_f)s + K_p K_m}$$

$$= \frac{10(10)}{(0.01)(2)s^3 + [(0.01)(0.25) + (6)(2)]s^2 + [(6)(0.25) + (10)(0.1) + (10)(7)] + (10)(10)}$$

$$\boxed{\frac{\theta_m}{\theta_{ref}} = \frac{100}{0.02s^3 + 12.0025s^2 + 72.5s + 100}} \Leftarrow (i)$$

zeroes: $\boxed{\text{none}}$

poles: $0.02s^3 + 12.0025s^2 + 72.5s + 100 = \emptyset$

$$\boxed{s = \{-594.037, -3.9657, -2.12245\}}$$

Given: A typical negative feedback system

a] $G_c(s) = 1,\quad G(s) = \dfrac{20s^3 + 60s^2 + 240s + 200}{s^4 + 5s^3 + 6s^2},\quad H(s) = \dfrac{2}{s+2}$

a] Routh Array to verify stability

$$A(s) = 1 + G_c(s)G(s)H(s) = 1 + \frac{(20s^3+60s^2+240s+200)\left(\frac{2}{s+2}\right)}{s^4+5s^3+6s^2} = 1 + \underbrace{\frac{(20s^3+60s^2+240s+200)\,2}{(s^4+5s^3+6s^2)(s+2)}}_{G_{open} = G(s)H(s)}$$

$$A(s) = \frac{s^5 + 7s^4 + 56s^3 + 132s^2 + 480s + 400}{s^5 + 7s^4 + 16s^3 + 12s^2}$$

RA: no sign changes for stability!!

s^5	1	56	480
s^4	7	132	400
s^3	37.14	422.86	Ø
s^2	52.3	400	Ø
s^1	138.81	Ø	
s^0	400		

$\boxed{\text{no sign changes } \therefore \text{ system is stable}}$

b] OL ⇒ $G(s)H(s) = \dfrac{2(20s^3+60s^2+240s+200)}{(s^4+5s^3+6s^2)(s+2)} \longrightarrow \;= (s+2)s^2(s^2+5s+6)$

Type 2 system

$$K_p = \lim_{s\to 0} \frac{2(20s^3+60s^2+240s+200)}{s^2(s+2)(s^2+5s+6)} = \frac{400}{\varnothing} = \infty \Rightarrow e_p(\infty) = \frac{1}{1+\infty} = \boxed{\varnothing}$$

$$K_v = \lim_{s\to 0} \cancel{s} \cdot \frac{2(20s^3+60s^2+240s+200)}{\cancel{s^2}(s+2)(s^2+5s+6)} = \frac{400}{\varnothing} = \infty \Rightarrow e_v(\infty) = \frac{1}{\infty} = \boxed{\varnothing}$$

$$K_a = \lim_{s\to 0} \cancel{s^2} \cdot \frac{2(20s^3+60s^2+240s+200)}{\cancel{s^2}(s+2)(s^2+5s+6)} = \frac{400}{12} = 33.3 \Rightarrow e_a(\infty) = \frac{1}{33.3} = \boxed{0.03}$$

3. **Given**: A typical negative feedback system, where $G(s)$ has three zeros at -1, $-1 \pm 2j$ and four poles at -3, -2, 0×2. Additionally, the sensor feedback is described by $H(s) = \frac{2}{s+2}$

Find: the Root Locus sketch and, if appropriate, include:

1. σ_i
2. Θ_k
3. BA and BI points
4. K at BA and BI

$-8/$

$$L(s) = \frac{k(s+1)(s+1+2j)(s+1-2j)(2}{(s+3)(s+2)(s)(s)\ (s+2}$$

k=1 for now

$j\omega$

Typical negative feedback system

$G(s)$ has:

zeroes: $\{-1, -1+2j, -1-2j\}$

poles: $\{-3, -2, \emptyset, \emptyset\}$

sensor feedback: $H(s) = \frac{2}{s+2}$

Find Root Locus and Sketch σ_i, Θ_k, BA and BI points, K at BA and BI

$$G(s)H(s) = \frac{2(s+1)(s+1+2j)(s+1-2j)}{(s+3)(s+2)^2 s^2}$$

zeroes: $\{-1, -1+2j, -1-2j\}$ $m = 3$

poles: $\{-3, -2, -2, \emptyset, \emptyset\}$ $n = 5$

$$\sigma_i = \frac{(-3)+(-2)+(-2)+\emptyset+\emptyset - [(-1)+(-1)+(-1)]}{5-3} = \boxed{-2}$$

$$K = \{\emptyset, 1\} \quad \Theta_k = \frac{(2K+1)\pi}{|n-m|} = \begin{cases} \Theta_0 = \frac{\pi}{2} \\ \Theta_1 = \frac{3\pi}{2} \end{cases}$$

BA: $K=1$, $N_L' D_L - D_L' N_L = \emptyset$

1. **Given** the system described by the diagram in Figure 1 where the transfer function G_{t} described by the diagram in Figure 2

Find:

A. Set $G_c = K$, find the Root Locus sketch with as much detail as possible (σ_i, θ_k, B BI, K at BA and BI)

B. Set $G_c = K_p + \frac{K_i}{s}$ and find (i) the range of K_p and K_i such that the system is stabl and (ii) select a loop-gain zero and sketch the Root Locus with as much detail a possible (σ_i, θ_k, BA, BI, K at BA and BI)

C. Set $G_c = K_p + \frac{K_i}{s} + K_d s$ and find (i) the range of K_p, K_i, and K_d such that the system is stable and (ii) select loop-gain zeros and sketch the Root Locus with as much detail as possible (σ_i, θ_k, BA, BI, K at BA and BI)

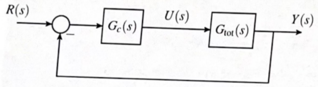


Figure 1: Block diagram of the controlled system.

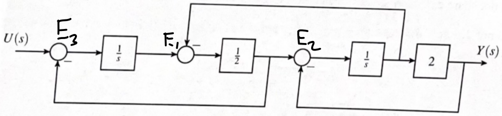


Figure 2: Block diagram of G_{tot}.

$$Y = \frac{2}{s} E_2$$

$$E_2 = \frac{1}{2} E_1 - \frac{2}{s} E_2$$

$$E_1 = \frac{1}{s} E_3 - \frac{1}{s} E_2$$

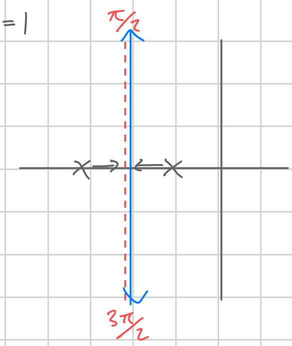
$$E_3 = U - E_1 \frac{1}{2} \Rightarrow$$

$$E_2\left(1 + \frac{2}{s}\right) = \frac{1}{2} E_1$$

$$E_2 = \frac{1}{2}\left(\frac{E_1}{1 + 2/s}\right)$$

$$L = G_c G H, \quad G_c = K, \quad G = \frac{1}{s^2 + 3s + 1}, \quad H = 1$$

$$L = \frac{K}{s^2 + 3s + 1} = \frac{K}{(s + 2.62)(s + 0.38)}$$



BA: $N_L' D_L - D_L' N_L$

$$\emptyset - (2s + 3)(1) = 0$$

$$s = -1.5$$

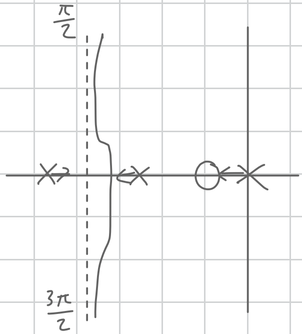
$$\sigma_i = \frac{(-2.62 - 0.38) - \emptyset}{2} = -1.5$$

$$K = \{0, 1\} \quad \theta_0 = \frac{\pi}{2}$$

$$\theta_1 = \frac{3\pi}{2}$$

b] $L = G_c G H$, $G_c = K_p + \frac{K_i}{s} = \frac{K_p(s+0.1)}{s}$, $G = \frac{1}{s^2+3s+1}$, $H=1$

$$\frac{K_p(s+0.1)}{s(s+2.62)(s+0.38)}$$



$$K_{BR} = \left.\frac{-D_L'}{N_L'}\right|_{s=BA} = 1.26$$

$$K_{CR} > \emptyset$$

$BA: N_L' D_L - D_L' N_L$

$$(1)(s)(s+2.62)(s+0.38) - (3s^3 + 6.03s^2 + 1.06s + 0.01) = \emptyset$$

$$s = -1.497$$

$$\sigma_i = \frac{(\emptyset - 2.62 - 0.32) - (-0.1)}{2} = -1.45$$

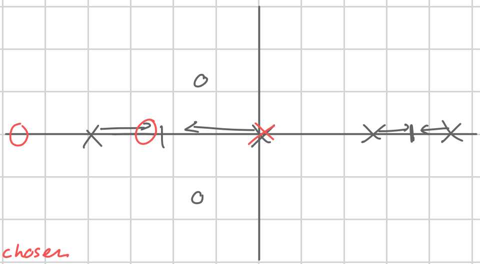
$K = \{0, 1\}$ $\quad \theta_0 = \frac{\pi}{2}$

$\theta_1 = \frac{3\pi}{2}$

c] $G =$

$$G(s) = \frac{s^2+2s+2}{(s^2-11s+24)(s^2+7s)} = \frac{s^2+2s+2}{s^4-4s^3-53s^2+168s} = \frac{(s+1+j)(s+1-j)}{s(s+7)(s-3)(s-8)}, \quad H=1$$

My chosen PID

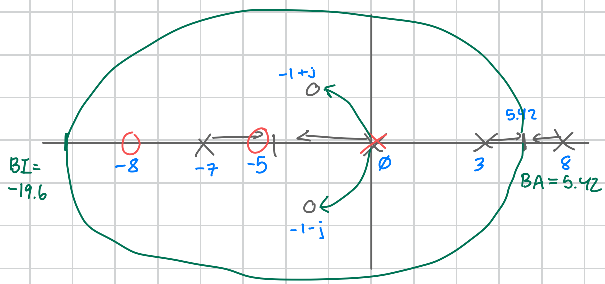


chosen

$$G_c = \frac{K_d(s+5)(s+8)}{s} \longrightarrow L = G_c G H = \frac{K_d(s+5)(s+8)(s+1+j)(s+1-j)}{s^2(s+7)(s-3)(s-8)} = \frac{K_d(s^4+15s^3+68s^2+106s+80)}{s^5-4s^4-53s^3+168s^2}$$

BA / BI: $N_L' D_L - D_L' N_L = \emptyset$

$$\underset{BA}{s = 5.42}, \quad \underset{BA}{\emptyset}, \quad \underset{BI}{-19.6}$$



$\sigma = 19$

$K = \{\emptyset\} \quad \Theta = \emptyset$

Given : $\frac{Y(s)}{R(s)} = \frac{s+126}{(s+6)(s^2+10s+21)}$

a] loop gain L, and steady state error for position, velocity, and acceleration

$$L = GH : \frac{1}{\frac{s^3+16s^2+81s+126}{s+126}} \longrightarrow \frac{1}{\frac{1}{G}+H}$$

$$H = \frac{126}{s+126}, \quad G = \frac{s+126}{s^3+16s^2+81s}$$

$$LG = GH = \left(\frac{126}{\cancel{s+126}}\right)\left(\frac{\cancel{s+126}}{s^3+16s^2+81s}\right)$$

$$\boxed{LG = \frac{126}{s(s^2+16s+81)}}$$

Type 1 system

i) $K_p = \lim_{s\to 0} \frac{126}{s(s^2+16s+81)} = \frac{126}{\emptyset} = \infty \Rightarrow e_p(\infty) = \frac{1}{1+\infty} = \boxed{\emptyset}$

ii) $K_v = \lim_{s\to 0} \cancel{s} \cdot \frac{126}{\cancel{s}(s^2+16s+81)} = \frac{126}{81} = 1.56 \Rightarrow e_v(\infty) = \frac{1}{1.56} = \boxed{0.64}$

iii) $K_a = \lim_{s\to 0} s^{\cancel{2}} \cdot \frac{126}{\cancel{s}(s^2+16s+81)} = \frac{\emptyset}{81} = \emptyset \Rightarrow e_a(\infty) = \frac{1}{\emptyset} = \boxed{\infty}$

b] when input is impulse, find (1) $y(\emptyset)$, (2) $y(\infty)$, (3) $y(1)$

$$Y(s) = \frac{s+126}{s^3+16s^2+81s+126} \overset{\text{expand}}{=} \frac{119}{4(s+7)} - \frac{40}{(s+6)} + \frac{41}{4(s+3)}$$

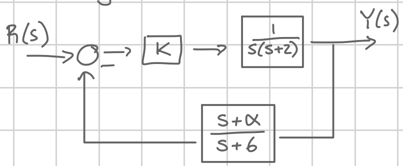
$$y(t) = \frac{119}{4}e^{-7t} - 40e^{-6t} + \frac{41}{4}e^{-3t}$$

(1) $y(\emptyset) = \lim_{t\to\infty}(1)[y(t)] = \emptyset - \emptyset + \emptyset = \boxed{\emptyset}$

(2) $y(\infty) = \lim_{t\to\emptyset}(1)[y(t)] = \frac{119}{4} - 40 + \frac{41}{4} = \boxed{\emptyset}$

(3) $y(1) = \frac{119}{4}e^{-7} - 40e^{-6} + \frac{41}{4}e^{-3} = \boxed{0.438}$

control system where $\varnothing < K < \infty$



a] Root Locus for system when $\alpha = 4$

$$T = \frac{\frac{K}{s(s+2)}}{1 + \frac{K}{s(s+2)}\left(\frac{s+4}{s+6}\right)} \cdot \frac{s(s+2)(s+6)}{s(s+2)(s+6)} = \frac{K(s+6)}{s(s+2)(s+6) + K(s+4)} = \frac{K(s+6)}{s^3+8s^2+12s+K(s+4)}$$

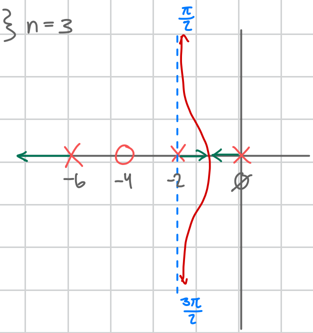
$$L = KGH = \frac{K(s+4)}{s(s+2)(s+6)}$$

zeroes: $\{-4\}$ $m=1$

poles: $\{\varnothing, -2, -6\}$ $n=3$

$$\sigma_i = \frac{(\varnothing-2-6)-(-4)}{3-1} = -2$$

$K = \{0, 1\}$ $\quad \Theta_k = \begin{cases} \Theta_\varnothing = \frac{\pi}{2} \\ \Theta_1 = \frac{3\pi}{2} \end{cases}$



BA: $K=1$, $N_L' D_L - D_L' N_L = \varnothing$

$$(1)(s^3+8s^2+12s) - (3s^2+16s+12)(s+4) = \varnothing$$

$s = -1.07$

b] Range of K where system stable w/ $\alpha = 4$

$$A = D_L + KN_L$$
$$= [s^3+8s^2+12s] + K[s+4]$$
$$= s^3+8s^2+(K+12)s+4K$$

$$\boxed{\text{system is stable when } K > \varnothing}$$

RA:

s^3	1	$(K+12)$
s^2	8	$4K$
s^1	$X_{[3,1]}$	$\varnothing$
s^0	$4K$	

$$X_{[3,1]} = \frac{(8)(K+12)-(4K)}{8} = \frac{4K-96}{8}$$

$\frac{4K-96}{8} > \varnothing$

$4K-96 > \varnothing$

$K > -24$

$4K > \varnothing$

$K > \varnothing$

c] Range of α where system is stable w/ $K=10$

$$A = D_L + KN_L$$
$$= [s^3+8s^2+12s] + [10][s+\alpha]$$
$$= s^3+8s^2+22s+10\alpha$$

$$\boxed{\text{system is stable when } \varnothing < \alpha < 17.6}$$

RA:

s^3	1	22
s^2	8	10α
s^1	$X_{[3,1]}$	$\varnothing$
s^0	10α	

$$X_{[3,1]} = \frac{(8)(22)-10\alpha}{8} = \frac{176-10\alpha}{8}$$

$\frac{176-10\alpha}{8} > \varnothing$

$176-10\alpha > \varnothing$

$\alpha < 17.6$

$10\alpha > \varnothing$

$\alpha > \varnothing$